Biology 1

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–12 · FULL YEAR

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. More investigations and examples are approved for this course than appear here, so scholars may see some variation in labs, activities, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Biology 1 introduces scholars to the study of living systems: their structure, function, and interconnection. As a foundational laboratory science, the course develops the practices of scientific investigation through careful observation, measurement, and experimentation — weekly hands-on data work is integral, not optional. In a VR- and AI-enhanced learning environment, scholars explore cellular structures, organisms, and ecosystems with the depth virtual environments make possible. Throughout, scholars build precision in observation, accuracy in interpreting and reporting evidence, responsibility in following scientific standards, and wonder at the complexity and beauty of the living world.

THE YEAR AT A GLANCE

1. **The Practice of Science** (foundation)
   Posing questions, designing investigations, gathering and interpreting data, and arguing from evidence.
2. **The Chemistry of Life** (strand)
   Macromolecules, the special properties of water, enzymes, ATP, and the paired engines of photosynthesis and cellular respiration.
3. **Cells** (strand)
   Cell theory and its discovery, plant and animal cells, membranes and transport, and the cell cycle — mitosis and meiosis compared.
4. **Genetics and Biotechnology** (strand)
   Mendel's laws and inheritance patterns, DNA replication and gene expression, mutation, and the promise and ethics of biotechnology.
5. **Evolution and Classification** (strand)
   The evidence for evolution, natural selection and other mechanisms of change, and how living things are classified.
6. **The Human Body and Ecology** (strand)
   Brain, heart, and immune system; human development; then outward to populations, food webs, biodiversity, and human impact on ecosystems.

A TYPICAL WEEK

- Lessons most days, with a lab or data investigation every week — the national standard for high-school science.
- A science notebook tracks observations, data tables, and evidence-based explanations.
- Immersive explorations on Optima's VR campus take scholars inside cells, organisms, and ecosystems.
- Lab reports and evidence-based writing develop scientific communication.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Lessons, labs, and notebook work fill most of the scheduled time.
- Lab-report and exam weeks may run longer.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A science notebook (digital or paper) for observations and data.
- Access to Optima's VR campus for immersive labs and explorations.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.